

EXPERIMENT No. 3

CO3: Implement the interfacing of sensors, actuators, and ICs, with microcontroller boards.

AIM: Interface of 16*2 Liquid crystal (LCD) with Arduino Uno

HARDWARE USED: - Arduino Uno, Breadboard, 16x2 LCD, 1-Resistor (1K Ω) resistors, Jumper wires.

THEORY:

To interface LCD to the Arduino requires Liquid Crystal library. The LCD module is used to print visible text on display, mainly this display is used for DIY project. The 16x2 LCD comes with 16 columns and 2 rows hence it's called as 16x2 LCD Module. In this display we can print maximum 32 characters. The LCD has many types like 8x1, 8x2, 10x2, 16x1, 20x4 etc. In this experiment we are using only 12 pins out of 16 pins. This display comes with blue or green backlight, with white or black text.

16X2 LIQUID CRYSTAL DISPLAY:

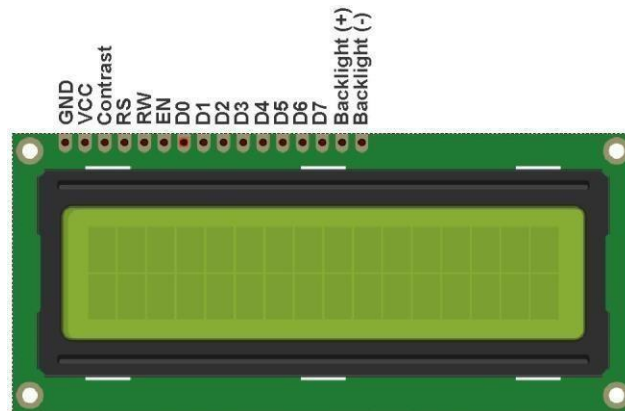


Figure 6: LCD (16*2)

The use of pins listed below:

- **GND(VSS):** Connect the ground pin of the power supply to this pin.
- **VCC:** Connect this pin to 5v
- **Contrast (VEE):** This pin is used to adjust the contrast of Display.
- **RS:** RS pin means Register select pin. Selects command register when the pin is LOW. And selects data register when this pin is HIGH.
- **RW:** It represents the Read Write pin. When this pin is LOW, the MCU writes to register. And when the pin is HIGH, MCU reads from the register. Here we want to write. It

connects it permanently to GND.

- **EN (E):** EN pin means the Enable pin. Send data to data pins when a HIGH to LOW pulse is given.
- **D0-D7 (DB0-DB7):** These are 8 data pins. Here we interface this LCD with Arduino in 4-bit mode. So, we need only D4 to D7.
- **Backlight (+):** This is the anode pin of the backlight of the display.
- **Backlight (-):** This is the cathode pin of the backlight of the display

CODE:

```
#include<LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

void setup()

{

  lcd.begin(16, 2);

}

void loop()

{

  lcd.setCursor(0,0);

  lcd.print(" CHANDIGARH");

  lcd.setCursor(3,1);

  lcd.print("UNIVERSITY");

}
```

Assembly

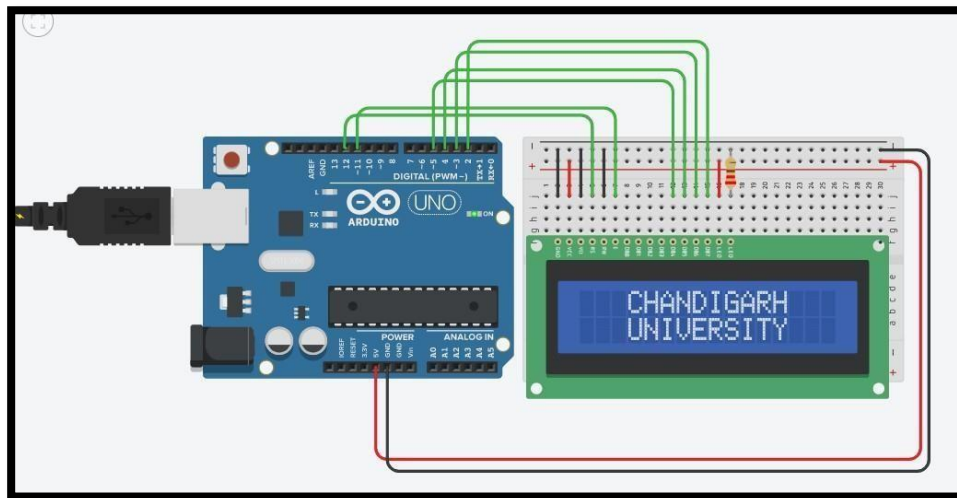


Figure 7: Interfacing of LCD with Arduino

PRECAUTIONS.

1. The proper resistance value must be chosen to avoid LCD burst.
2. Connections should not be loose

REVIEW QUESTIONS:

S. No.	Question
1.	Describe the role of potentiometer in LCD.
2.	State the use of <code>lcd.begin()</code> .
3.	State the importance of <code>lcd.setCursor()</code> while interfacing.
4.	Write the library included in case of LCD interfacing?
5.	Is it possible to draw a custom character in case of LCD interfacing.